MILESTONE 1: Model Selection and Architecture – LegalEase

Focus: selecting the optimal generative AI model, defining the system architecture, and setting up the development environment.

Activity 1.1: Research and Select the Appropriate Generative AI Model

1. Understand the Project Requirements: LegalEase must generate structured legal documents such as contracts, NDAs, and agreements based on document type, parties involved, terms and conditions, and effective dates.

2. Explore Gemini Model Documentation: Review available Gemini models and the capabilities of gemini-1.5-pro, including long context window, structured prompting, fast generation speed, and legal-domain compatibility.

3. Evaluate Model Performance: The source states that gemini-1.5-pro is capable of handling complex formatting, reasoning, and structured text generation.

4. Select the Optimal Model: Selected: gemini-1.5-pro. It is described as suited for formal documents and dynamically generating clauses from user inputs.

Activity 1.2: Define the Architecture of the Application

- Frontend: Streamlit for user interaction, document input, preview, editing, and downloads.
- Backend: FastAPI app to route input to Gemini and return generated content.
- AI Core: Gemini integration for content generation.
- Formatting Modules: DOCX, PDF, TXT generators.

Frontend Functionality

- TextInput for document type and dates.
- TextArea for parties and terms.
- Buttons: Generate Document, Edit Document, Download as TXT / DOCX / PDF.

Backend Responsibilities

- Expose POST /generate endpoint via FastAPI.
- Accept JSON input: document_type, parties, terms, dates.
- Route to GeminiDocumentGenerator and return structured document text.

AI Integration Points

GeminiDocumentGenerator forms a structured prompt and sends it to Gemini via generate_content; the generated text is returned to FastAPI, then displayed and made available for editing/exporting.

Activity 1.3: Set Up the Development Environment

- Install Python 3.10+ and pip.
- Create a virtual environment.
- Install FastAPI and Uvicorn.
- Install Streamlit.
- Install Google Generative AI SDK.
- Install python-docx, fpdf, Pillow, requests, and python-dotenv.
- Set up the application structure.